5A - 6A - 10A - Atmosphere - EvolLand

Test 3 Master Test Review

50 Review Questions

STUDENT EDITION

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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REVIEW

Test 3 Master Test Review

50 questions worth focusing on as you prepare for Test 3, organized by subject. Each one tells you where to find the answer. Work through them in your own words, typing your answers on the lines.

The Earth’s Atmosphere

1. What is the atmosphere? Excluding water vapor, name the four main gases that make it up, with their percentages.

Find answers in: Module 5A, 5AE1

2. Name the four layers of the atmosphere in order, from the ground up, and the “pause” at the top of each. What is a pause?

Find answers in: Module 5A, 5AE1

3. In which two layers does the temperature DECREASE with increasing altitude, and in which two does it INCREASE? What heats each of the two warming layers?

Find answers in: Module 5A, 5AE1

4. What makes the troposphere special, even though it is the thinnest layer? About how high is its top at mid-latitudes, and why is it taller at the equator than at the poles?

Find answers in: Module 5A, 5AE1

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5. What are jet streams? In the Northern Hemisphere, where are the two jet streams found, and why do airliners like to ride them?

Find answers in: Module 5A, 5AE1

6. In which layer is the ozone layer, and what does it do? Which two gases in that layer protect life from deadly UV radiation? Why is ozone “good up high, bad nearby”?

Find answers in: Module 5A, 5AE1

7. What is the ionosphere, and where is it? Up to what frequency does it reflect radio waves, and what is “skip”?

Find answers in: Module 5A, 5AE1

8. The thermosphere can reach 2000° C. Why wouldn’t an astronaut out there feel hot? What is the difference between temperature and heat?

Find answers in: Module 5A, 5AE1, 5AE2

9. What causes the aurorae, and in which layer do they occur? What “sphere” harmlessly traps the charged particles of the solar wind?

Find answers in: Module 5A, 5AE1

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10. What is Earth-based weather? Name the six main ingredients of Earth-based weather.

Find answers in: Module 5A, 5AE2

11. Name four forms of precipitation and describe each. How do hailstones grow? Are fog, dew, and clouds forms of precipitation?

Find answers in: Module 5A, 5AE2

12. What is lightning, and what is thunder? You count 10 seconds between a flash and its thunder: about how far away was the strike? What are summer cumulonimbus clouds made of?

Find answers in: Module 5A, 5AE2

13. What is a greenhouse gas? Name three. Why are nitrogen and oxygen NOT greenhouse gases?

Find answers in: Module 5A, 5AE2

14. What is space weather? What are the solar wind and a coronal mass ejection (CME), how long does each take to reach Earth, and how do satellites at L1 give us warning?

Find answers in: Module 5A, 5AE2

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15. What makes high and low pressure, and what weather does each bring? In the Northern Hemisphere, which way does air circle a high and a low, and why?

Find answers in: Module 5A, 5AE2 (Homework #11)

Atmospheric Pressure Exercise

16. Define pressure and atmospheric pressure. How does atmospheric pressure change as you go higher, and why?

Find answers in: Atmospheric Pressure exercise

17. Who invented the mercury barometer, and when? How tall is the mercury column at sea level, and what holds it up? What is the general relationship between pressure and weather?

Find answers in: Atmospheric Pressure exercise

18. Give the value of one Atmosphere in Pascals, millibars (hectopascals), psi, mm Hg, and in Hg. In what unit is blood pressure reported?

Find answers in: Atmospheric Pressure exercise

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Ancient Atmospheres

19. What happened to Earth's first (primordial) atmosphere? Where did the second atmosphere come from, and what gases was it mostly made of?

Find answers in: Ancient Atmospheres video (Module 5A, Homework #6)

20. Once early Earth cooled below the boiling point of water, what happened, for about how long, and what was the result?

Find answers in: Ancient Atmospheres video (Module 5A, Homework #6)

21. What was the Great Oxygenation Event? Which organisms caused it, when did it start, and why did the oxygen take so long to reach the air?

Find answers in: Ancient Atmospheres video (Module 5A, Homework #6)

EvolLand Exercise

22. About how many millions of years ago was the Cambrian Explosion, and what happened in it?

Find answers in: EvolLand exercise

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23. As oxygen in the atmosphere rose, life moved from the sea to the land. About how high did oxygen peak, and when? What happened to flying insects?

Find answers in: EvolLand exercise (Module 5A, Homework #6)

24. In years, about how long ago did our species, Homo sapiens, evolve?

Find answers in: EvolLand exercise

The Nature of Light

25. What is optics? Give the simple definition of light, and the more complex (but limited) definition.

Find answers in: Module 6A, 6AE1

26. In the definition of light, what do “dual transverse,” “self-propagating,” and “non-mechanical” mean?

Find answers in: Module 6A, 6AE1

27. What is the speed of light in a vacuum, exactly and rounded, in m/s, km/s, and miles/s? Why is the exact value exact?

Find answers in: Module 6A, 6AE1

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28. Who first used astronomy to determine the speed of light, when, and how?

Find answers in: Module 6A

29. Is light a wave or a particle? Why is light's nature called contradictory, and why does that make even the complex definition of light "limited"?

Find answers in: Module 6A, 6AE1

30. What are the two schools of light? For each, give its founder, what he said light is made of, and how the school draws light's path.

Find answers in: Module 6A, 6AE1

31. What did James Clerk Maxwell unify, what did his equations predict, and what did he conclude about light?

Find answers in: Module 6A, 6AE1

32. State the Inverse Square Law. If you move three times as far from a lamp that shines in all directions, what fraction of the original intensity reaches you?

Find answers in: Module 6A, 6AE1

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33. What do “temporally coherent” and “spatially coherent” mean? Why is laser light coherent, and why does it dim so slowly with distance?

Find answers in: Module 6A, 6AE1

34. List the electromagnetic spectrum in order of increasing frequency. About what range of wavelengths can humans see? Are cosmic rays part of the spectrum?

Find answers in: Module 6A, 6AE1

35. What are the three ways light interacts with matter? Define each.

Find answers in: Module 6A, 6AE1

36. Name and define the five classical phenomena of light. Which model of light explains all five, and which one can the particle model adequately explain?

Find answers in: Module 6A, 6AE1

37. What is the difference between specular and diffuse reflection? Do all shiny objects exhibit specular reflection?

Find answers in: Module 6A

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38. What causes refraction? When light passes from air into water, does it speed up or slow down, and does the ray bend toward or away from the normal?

Find answers in: Module 6A

39. True or false: Cherenkov radiation is the blue glow given off when charged particles, such as electrons, travel through water faster than light itself travels in water.

Find answers in: Module 6A

The Exploration of Space

40. What was Sputnik 1? Who launched it, and when? Why could no one doubt it was real?

Find answers in: Module 10A

41. What was the first satellite the United States successfully launched, and when? What did it discover?

Find answers in: Module 10A

42. What made Telstar 1 historic, and what ended its life?

Find answers in: Module 10A

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43. Who was the first person to orbit the Earth? When, and in what spacecraft? How long did the flight take, and how did he land?

Find answers in: Module 10A

44. Who were the first and second Americans in space? Name their capsules. Why was neither of them the first American to orbit the Earth?

Find answers in: Module 10A

45. Who was the first American to orbit the Earth? When, in what capsule, and for how many orbits? What scare did he face on reentry?

Find answers in: Module 10A

46. What were the two parts of President Kennedy's 1961 challenge? Which Apollo 11 astronauts first stepped on the Moon, when, and where did they land? Who stayed in orbit?

Find answers in: Module 10A

47. How many Apollo missions set out to land on the Moon, how many succeeded, and how many men have walked on the Moon?

Find answers in: Module 10A

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48. Name the four pioneers of earthbound flight in the First Extended Study, and give one "first" for each.

Find answers in: Module 10A

49. Which uncrewed missions produced the first close-up images of Jupiter and of Saturn? What else was first about the earlier one?

Find answers in: Module 10A

50. What were the Viking missions, and what was their key first? Which spacecraft is the only one to have visited Uranus and Neptune?

Find answers in: Module 10A

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